# Vishal Kumar Allam

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## Professional Summary

- Results-driven **Software Engineer / Palantir Data Engineer** with **5+ years of experience** designing, developing, integrating, and supporting enterprise data platforms, analytics solutions, operational applications, and AI-enabled systems across healthcare, SaaS, and product environments.
- Hands-on expertise with **Palantir Foundry** including Workshop, Ontology, Pipeline Builder, Code Repositories, Functions, Actions, data products, transformations, operational applications, and reusable enterprise workflows.
- Strong data engineering experience using **Python, PySpark, SQL, Apache Spark, Databricks, Azure Data Factory, Azure Data Lake, Microsoft Fabric, AWS, and REST APIs** for scalable data processing, analytics, and enterprise integration.
- Experienced designing **Ontology models, business objects, relationships, actions, knowledge graphs, and operational workflows** that support reporting, decision support, advanced analytics, AI/ML, and semantic data access.
- Expertise implementing **data quality, validation, testing, metadata management, data lineage, governance, role-based access control, and DevSecOps** practices to create trusted and reusable enterprise data assets.
- Hands-on experience with **Palantir AIP, AI/ML, NLP, semantic search, knowledge graphs, LLM-enabled analytics, and intelligent decision-support solutions** integrated with enterprise data ecosystems.
- Healthcare technology experience involving health data, HL7, clinical and laboratory data, payer workflows, healthcare analytics, public health use cases, and secure data processing in multidisciplinary Agile environments.

## Education

**University of Missouri Kansas City** — Kansas City, MO
*"Master of Science" in Computer Science* — *Jan. 2023 – May 2024*

**Jawaharlal Nehru Technological University** — Hyderabad, India
*"Bachelor of Technology" in Computer Science* — *Aug. 2015 - May 2019*

## Technical Skills

- **Palantir Technologies:** Palantir Foundry, Palantir AIP, Workshop, Ontology, Pipeline Builder, Functions, Actions, Code Repositories, Data Products, Operational Applications
- **Programming & Data Processing:** Python, PySpark, SQL, Apache Spark, Java, TypeScript, JavaScript, Pandas, NumPy, Large-Scale Data Processing
- **Data Engineering:** ETL/ELT Pipelines, Data Transformations, Data Modeling, Structured/Unstructured Data, Data Products, Data Integration, Data Quality, Validation, Metadata Management, Data Lineage
- **Cloud & Enterprise Platforms:** Azure, AWS, Google Cloud, Databricks, Azure Data Factory, Azure Data Lake, Microsoft Fabric, S3, REST APIs, Enterprise Databases
- **AI & Advanced Analytics:** Palantir AIP, Machine Learning, NLP, Knowledge Graphs, Semantic Search, LLMs, Generative AI, AI-Enabled Analytics, Decision Support
- **Governance & Security:** Enterprise Data Governance, RBAC, Security/Access Management, Metadata Governance, Data Lineage, DevSecOps, Git, CI/CD, Automated Testing
- **Healthcare & Delivery:** HL7, Healthcare Data, Clinical Data, Laboratory Data, Public Health Analytics, Agile Scrum, Jira, Confluence, System Diagrams, Data Flow Diagrams, SOPs, Implementation Guides

## Experience

**Software Engineer AI** Oct 2024 – Present
***PCMI LLC (Full Time)*** *Charlotte, NC, USA*

- Designed and developed enterprise data and operational solutions using **Palantir Foundry**, including Workshop applications, Ontology models, Pipeline Builder workflows, Functions, Actions, and reusable data products supporting reporting, analytics, AI, and decision-support use cases.
- Built scalable data pipelines and transformation workflows using **Python, PySpark, SQL, Apache Spark, and Foundry Code Repositories** to process structured and unstructured data from enterprise applications, APIs, relational databases, files, and cloud platforms.
- Designed and maintained **Ontology business objects, relationships, actions, and operational workflows**, translating complex business concepts into reusable semantic models supporting user-facing applications and analytical use cases.
- Developed interactive **Foundry Workshop** applications that exposed operational data, analytics, workflow actions, and decision-support capabilities to business users through ontology-backed interfaces.
- Implemented comprehensive data quality frameworks covering validation rules, schema checks, reconciliation, metadata management, lineage, governance, and role-based access controls to maintain trusted enterprise data assets.
- Integrated Foundry with **AWS, Azure, Databricks, Apache Spark, REST APIs, relational databases, and external data sources**, applying scalable and secure enterprise integration patterns.
- Integrated **Palantir AIP, LLMs, NLP, semantic search, knowledge graphs, and AI-enabled analytics** into operational workflows to support intelligent search, summarization, recommendations, automated analysis, and business decision support.
- Collaborated with data engineers, architects, data scientists, analysts, Product Owners, and business stakeholders during Agile sprint planning, backlog refinement, testing, deployment, troubleshooting, and continuous improvement while preparing architecture diagrams, data-flow diagrams, SOPs, and implementation guides.

**Environment:** Palantir Foundry, Palantir AIP, Workshop, Ontology, Pipeline Builder, Functions, Actions, Code Repositories, Python, PySpark, SQL, Apache Spark, Databricks, AWS, Azure, REST APIs, Data Products, Data Modeling, Data Quality, Metadata Management, Data Lineage, Governance, RBAC, NLP, Knowledge Graphs, Semantic Search, LLMs, Git, CI/CD, DevSecOps, Agile Scrum

**Software Engineer AI** Dec 2023 – Sept 2024
***Client: CVS*** *Remote, USA*

- Designed healthcare-focused data integration and analytics solutions using **Palantir Foundry**, building scalable pipelines, transformations, reusable data products, Ontology models, and operational applications supporting healthcare reporting and decision support.
- Developed data pipelines using **Python, PySpark, SQL, Apache Spark, and Databricks** to ingest and transform structured and unstructured healthcare data from APIs, enterprise databases, files, clinical platforms, and cloud environments.
- Built and maintained **Ontology models** representing patients, providers, encounters, claims, clinical events, and related healthcare entities, defining relationships and actions that enabled analytical and operational workflows.
- Developed user-facing **Foundry Workshop applications** to provide healthcare users with operational dashboards, analytical views, workflow actions, data exploration capabilities, and AI-assisted decision support.
- Implemented data quality, validation, testing, reconciliation, metadata management, lineage, governance, and RBAC practices to improve reliability, traceability, security, and reuse of healthcare data assets.
- Integrated Palantir with **Azure Data Factory, Azure Data Lake, Microsoft Fabric, AWS services, Databricks, databases, REST APIs, and external data sources** using scalable enterprise integration patterns.
- Supported **Palantir AIP, AI/ML, NLP, semantic search, knowledge graphs, and LLM-enabled analytics** for intelligent healthcare use cases involving operational analysis, data discovery, summarization, and decision support.
- Worked with healthcare data including **HL7, clinical data, laboratory data, payer information, and health-related datasets**, collaborating with multidisciplinary engineering, analytics, Product, and business teams through Agile and DevSecOps delivery practices.

**Environment:** Palantir Foundry, Palantir AIP, Workshop, Ontology, Pipeline Builder, Python, PySpark, SQL, Apache Spark, Databricks, Azure Data Factory, Azure Data Lake, Microsoft Fabric, AWS, REST APIs, HL7, Healthcare Data, Clinical Data, Data Products, Data Quality, Metadata Management, Data Lineage, Governance, RBAC, AI/ML, NLP, Semantic Search, Knowledge Graphs, Git, CI/CD, DevSecOps

**Software Engineer** Sept 2021 – Dec 2022
***Infor*** *Hyderabad, India*

- Developed enterprise data processing and analytics solutions using **Python, SQL, PySpark, and Apache Spark**, supporting large-scale ingestion, transformation, validation, and downstream application workflows.
- Designed reusable ETL/ELT pipelines integrating structured and unstructured data from enterprise databases, files, REST APIs, and application platforms while maintaining scalable transformation logic and reusable data models.
- Created relational and analytical data models, transformation rules, schema mappings, and reusable data products supporting operational reporting, analytical processing, and enterprise application requirements.
- Implemented data quality checks, reconciliation workflows, validation rules, metadata documentation, and lineage tracking to improve reliability and traceability across enterprise datasets.
- Worked with cloud-hosted data environments and enterprise data architectures involving AWS/Azure services, APIs, databases, and distributed processing components.
- Developed Python and SQL utilities for data extraction, transformation, validation, query optimization, and performance troubleshooting across high-volume enterprise datasets.
- Integrated automated validation and data-processing workflows with Git and Jenkins-based CI/CD pipelines, supporting Agile development, testing, deployment, troubleshooting, and continuous improvement.
- Collaborated with engineers, architects, analysts, Product Managers, and business stakeholders to translate requirements into scalable technical designs and produced solution documentation, data-flow diagrams, implementation notes, and operational procedures.

**Environment:** Python, PySpark, SQL, Apache Spark, ETL/ELT, REST APIs, Data Modeling, Data Transformations, Data Quality, Metadata Management, Data Lineage, AWS, Azure, Oracle, MySQL, Git, Jenkins, CI/CD, Agile Scrum, Technical Documentation

**Associate Software Engineer** June 2019 – August 2021
***Virtusa*** *Hyderabad, India*

- Supported enterprise data integration and software development initiatives using Java, Python, SQL, REST APIs, and relational databases across multi-tier enterprise applications.
- Developed data-processing and integration workflows to ingest, transform, validate, and exchange structured data between enterprise applications, backend services, databases, and external interfaces.
- Created SQL-based data validation, reconciliation, and transformation logic to ensure accuracy, consistency, and completeness of enterprise datasets across operational workflows.
- Supported data modeling, schema mapping, database integration, and API-driven application workflows while troubleshooting data and system issues across distributed application components.
- Implemented automated testing and validation using Selenium WebDriver, TestNG, Cucumber BDD, API testing, and SQL to ensure reliability of integrated enterprise applications and data flows.
- Participated in Git, Jenkins, Maven, and CI/CD-based delivery processes, supporting automated builds, testing, deployments, configuration management, and Agile software development.
- Collaborated with engineers, business analysts, QA leads, and Product stakeholders to analyze requirements, troubleshoot integration issues, identify technical risks, and support production releases.
- Created technical documentation including data-validation results, integration specifications, application workflows, troubleshooting procedures, test documentation, and production-support notes.

**Environment:** Java, Python, SQL, REST APIs, Enterprise Data Integration, Data Transformation, Data Validation, Relational Databases, Git, Jenkins, Maven, CI/CD, Agile Scrum

## Awards & Achievements

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded "Software Process Achievement" Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.